

Waguea Carine Fongang

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Education

University of Pennsylvania

B.S.E. in Computer Engineering

Philadelphia, PA

Expected May 2029

- **Relevant Coursework:** Data Structures & Algorithms, Programming Languages & Techniques I/II, Mathematical Foundations of Computer Science, Introduction to Computer Systems, Electrical Circuits and Systems
- **Honors & Programs:** Africanist Award (African Leadership Academy); AI4ALL Ignite Fellow; CodePath Technical Interview Prep Fellow

Technical Skills

Languages: C, C++, Verilog, Python, Bash, OCaml, SQL, R

Systems & Tools: Linux/Unix, LSF, Snakemake, Docker, Git, Windows API, WebView2, Google Cloud, Firebase

Experience

Logsdon Lab, University of Pennsylvania

Research Assistant

June 2026 – August 2026

Philadelphia, PA

- Built an automated **Python/Bash pipeline** to measure disease-linked DNA repeat copy number across **139 diverse human genome assemblies**, including the first characterization of understudied African populations, orchestrating parallel analyses on an HPC cluster using LSF and Snakemake.
- Developed **RepeatContact**, a Python signal-processing tool that counts DNA repeats directly from raw Nanopore electrical signals using Dynamic Time Warping and Fourier analysis; evaluated its behavior using simulated genomic data.
- Identified a **31% systematic discrepancy** between two DNA repeat-counting methods; implemented a third, independently anchored validation approach to investigate the difference.

Netter Center for Community Partnerships, University of Pennsylvania

Website Support Associate

April 2026 – Present

Philadelphia, PA

- Maintain **three production websites** supporting a network of **300 universities and 1,200+ community partners**, managing technical updates and public-facing content.

JoyNet – SAFELab, University of Pennsylvania

Software Engineer (Fall 2026)

September 2026 – December 2026

Remote

- Part-time Software Engineer on a social platform for positive short-form video, part of SAFELab at UPenn, working directly alongside the founder on day-to-day product development across the stack.

Projects

FilePeek | C++, Python, Windows API, NVIDIA NemoTron

August 2026

- Built a **native C++ Windows application** integrating with File Explorer to generate AI-powered summaries across **8 file formats**, allowing users to inspect files without opening them.
- Designed an asynchronous C++/Python architecture with file-aware caching; reduced average cold-summary latency by **28.6%** (**12.92s to 9.22s**).

Wonderly – HackMIT 2026 | Python, OpenCV, Computer Vision

September 2026

- Co-developed an **AI-powered art-education prototype**; led **computer vision development** with image capture, scene scanning, blur detection, duplicate-frame filtering, and edge/contour extraction.

Student Involvement

Wharton Undergraduate Data Analytics Club | Rewriting the Code | Computer Science Society | National Society of Black Engineers (NSBE) | ColorStack